01 Inputs

Judge model produces the extracted answer, reason, and label.

Ground-truth Answer
Dataset · standard answer

+

Last Prediction
Training model · final output

+

Extracted Answer + Judge Reason
Judge model · extracted evidence

+

Judge Label
Judge model · True / False

02 Judge Audit Skill

Reconstruct · verify correctness · audit extraction and label.

1 **Reconstruct**
Model's final answer

→

2 **Verify Training Model**
Whether the training model is actually correct

3 **Audit Judge Reliability**
Extraction and label reliability

4 **Output Audit Result**
Audit result and structured evidence

Judge_OK
Whether the judge label is correct · 1 = correct · 0 = incorrect

`judge_ok` `error type` `evidence` `possible new pattern`

`Judge_OK = 0 → Self-Evolution`
`Judge_OK = 1 → Statistics only`

Incorrect cases enter self-evolution; correct cases need no further action.

03 Self-Evolution

Collect failures · update the audit skill · improve the next round.

Failure Case Pool
Collect failure cases · store high-risk cases

→

Audit Skill Update
Mine patterns · refine rules/prompts · add examples

→

Stronger Audit in Next Round
Apply the updated skill · improve audit quality

FEEDBACK LOOP · UPDATED AUDIT SKILL FEEDS THE NEXT AUDIT ROUND